

Video Editor Mini App

Duration: ~2 hours · Confidential

— CONTEXT

Our team builds internal tools for content creators. We need a prototype that lets users clip, arrange, and export segments from YouTube videos — running reliably on constrained cloud infrastructure.

— WHAT WE NEED

Build a web application where a user can:

1. **Input a YouTube video URL** — the system downloads and makes it available for editing
2. **Select crop positions** — choose one or more time ranges from the video
3. **Merge clips** — combine selected segments into a single output video
4. **Apply transition effects** — add transitions between merged segments (e.g. fade, cut, slide)

The experience should feel complete: paste a link, select clips, export, download result.

— CONSTRAINTS

Time budget: ~2 hours

Stack preference: NestJS, React, and/or Python — but you decide the architecture

Infrastructure target: the system must be designed to run on AWS ECS Fargate with **0.5 vCPU and 1GB RAM**. You can simulate this locally with Docker: `--memory=1g --cpus=0.5`

Must be runnable by our team — Docker preferred, but any reproducible setup is acceptable

— WHAT WE'LL EVALUATE

We're not looking for a specific implementation. We're looking at how you think:

- **System design:** How did you break down the problem? What tradeoffs did you make?
- **Resource management:** How does your system behave under the memory/CPU constraint? What would break first, and how did you design around it?
- **Code quality:** Is the code readable, structured, and maintainable?
- **Product sense:** Does the result actually work as a user experience?
- **Engineering judgment:** What did you choose *not* to build, and why?

Please include a short write-up (in README or a separate doc) explaining your design decisions. We care as much about your reasoning as the code itself.

— OPEN QUESTION — SCALING

If 1,000 users submitted videos simultaneously, what would break first in your system — and how would you fix it?

Write your answer in the README. There's no right answer — we're evaluating your reasoning, not the idea itself.